$r(t) = s_i(t) + n(t)$ → [ kF ]

$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\sqrt{\frac{E_b}{2N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right)$$

$$E_d = \int_0^T [s_1(t) - s_2(t)]^2 dt$$

$$\rho = \frac{1}{E_b}\int_0^T s_1(t)\, s_2(t)\, dt$$

$E_b$ → ortalama bit enerjisi (average energy per bit)

$\rho$ → karşı ilişki katsayısı $-1 \le \rho \le 1$

$\gamma_b = \frac{E_b}{N_0}$ : Bit başına İGO (SNR per bit)

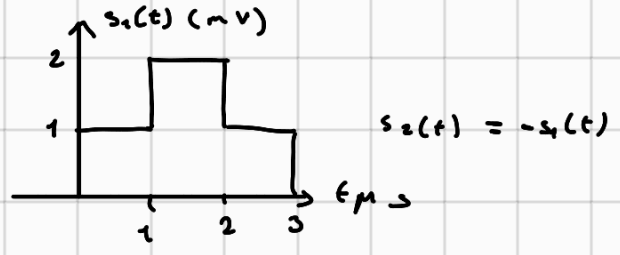
$s_1(t) \perp s_2(t)$ ($s_1(t)$ ve $s_2(t)$ ortogonal) $\Rightarrow \rho = 0 \Rightarrow P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$

$s_1(t) = -s_2(t)$ ( " " " zıt işaretler) (antipodal) $\Rightarrow \rho = -1 \Rightarrow P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$

$$E_b = \sum_{i=1}^{2} E_i P(s_i) = E_1 P(s_1) + E_2 P(s_2)$$

($E_i$ : $s_i(t)$'nin enerjisi)

**Ex:** Consider a binary communication system that receives equally likely signals $s_1(t)$ and $s_2(t)$ plus AWGN. Assume that the receiving filter is a MF, and the noise PSD $N_0$ is $10^{-12}$ Watt/Hz. Use the values of the received signal voltage and time shown in the figure below to compute bit error probability.



$P(s_1) = P(s_2) = \frac{1}{2}$

$N_0 = 10^{-12}\ W$

$s_2 = -s_1 \rightarrow E_b = E_{s_1} = E_{s_2}$

$$E_b = E_{s_1} = \int_0^T s_1^2(t) = \int_0^{10^{-6}} (10^{-3})^2 dt + \int_{10^{-6}}^{2\cdot10^{-6}} (2\cdot10^{-3})^2 dt + \int_{2\cdot10^{-6}}^{3\cdot10^{-6}} (10^{-3})^2 dt = 6\cdot10^{-12}\ J$$

$$P_b = Q\left(\sqrt{\frac{2\cdot6\cdot10^{-12}}{10^{-12}}}\right) = Q(\sqrt{12}) = Q(3{,}46) = 3\cdot10^{-4} \quad (\text{Tablodan})$$

$$= \frac{3}{10\,000}$$

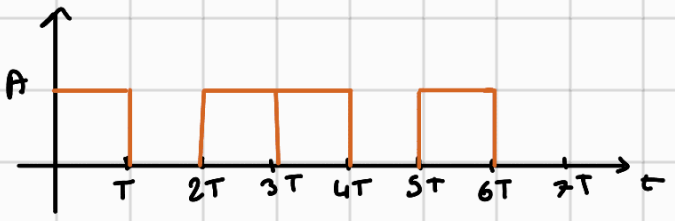
## İkili işaretleşmeni Hata olasılığı Performansı

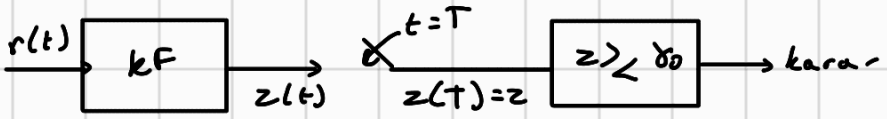
### Tek kutuplu işaretleşme (Unipolar signaling)

"1" $\rightarrow s_1(t) = A,\ 0 \le t \le T,\ A > 0$

"0" $\rightarrow s_2(t) = 0,\ 0 \le t \le T$

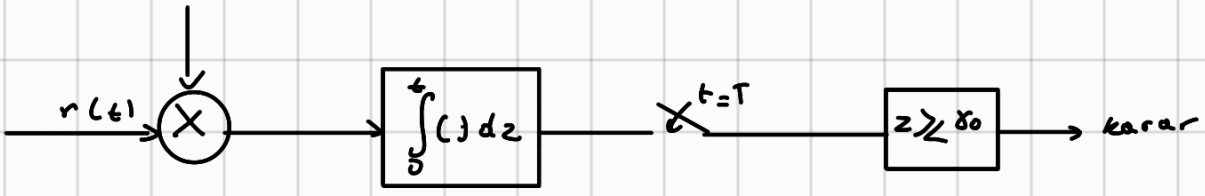
$$\Rightarrow \rho = \frac{1}{E_b} \int_0^T s_1(t)\, s_2(t)\, dt = 0$$




$$h(t) = s_1(T-t) - s_2(T-t) = s_1(T-t),\ 0 \le t \le T$$

$$s_1(t) - s_2(t) = s_1(t) = A,\ 0 \le t \le T$$




$$r(t) = s_i(t) + n(t)$$

$$z(t) = r(t) \circledast h(t) \xrightarrow{t=T} z = z(T) = \underbrace{r(t) \circledast h(t)}_{a_i(t) + n_0(t)} \Big|_{t=T}$$

$$\Rightarrow z = z(T) = a_i(T) + n_0(T) = a_i + n_0$$

$$a_i = a_i(T) = s_i(t) \circledast h(t)\Big|_{t=T}, \quad \boxed{i = 1,\ 2}$$

$$a_i = a_i(T) = \int_0^T s_i(t)\,[s_1(t) - s_2(t)]\, dt \qquad i = 1, 2$$

Tek katsayı işaretleşme $\Rightarrow a_i = \int_0^T s_i(t)\,[s_1(t) - \underbrace{s_2(t)}_{0}]\,dt \qquad i = 1,2$

$$a_1 = \int_0^T s_1^2(t)\,dt = \int_0^T A^2\,dt \qquad \gamma_0 = \frac{a_1 + a_2}{2} = \frac{A^2 T}{2}$$
$$a_2 = \int_0^T s_2(t)\,s_1(t)\,dt = 0$$

Eğer $z > \gamma_0 = \frac{A^2 T}{2} \Rightarrow$ karar $= s_1(t)$

$z < \gamma_0 \Rightarrow$ karar $= s_2(t)$

$$E_d = \int_0^T [s_1(t) - s_2(t)]^2\,dt = \int_0^T s_1^2(t)\,dt = A^2 T$$

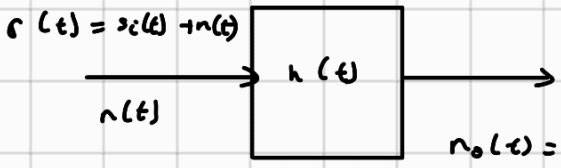
$$P_b = Q\left(\sqrt{\frac{E_d}{2N_0}}\right) = Q\left(\sqrt{\frac{A^2 T}{2N_0}}\right) \Rightarrow E_b = E_1 \underbrace{P(s_1)}_{1/2} + E_2 \underbrace{P(s_2)}_{1/2} = \frac{A^2 T}{2}$$

$$E_1 = \int_0^T s_1^2(t)\,dt = A^2 T$$

$$P_b = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$P_b = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right) = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right)$$



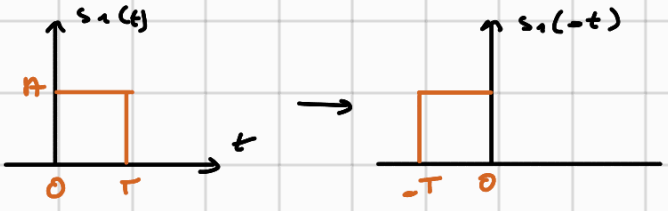

$n_0(t) = n(t) \otimes h(t)$

$S_n(f) = \frac{N_0}{2}$ Watt/Hz

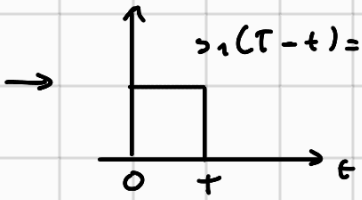
$S_{no}(f) = S_n(f)\,|H(f)|^2$

$$\sigma_0^2 = P_{no} = \int_{-\infty}^{\infty} S_{no}(f)\,df = \frac{N_0}{2}\underbrace{\int_{-\infty}^{\infty} |H(f)|^2\,df}_{E_h} = \frac{N_0}{2} E_h$$

$$\boxed{\sigma_0^2 = \frac{N_0}{2} E_h}$$




Tek katsayılı da $h(t) = s_1(T-t)$




$h(t) = s_1(t) \Rightarrow E_h = E_{s_1} = E_1 = A^2 T$

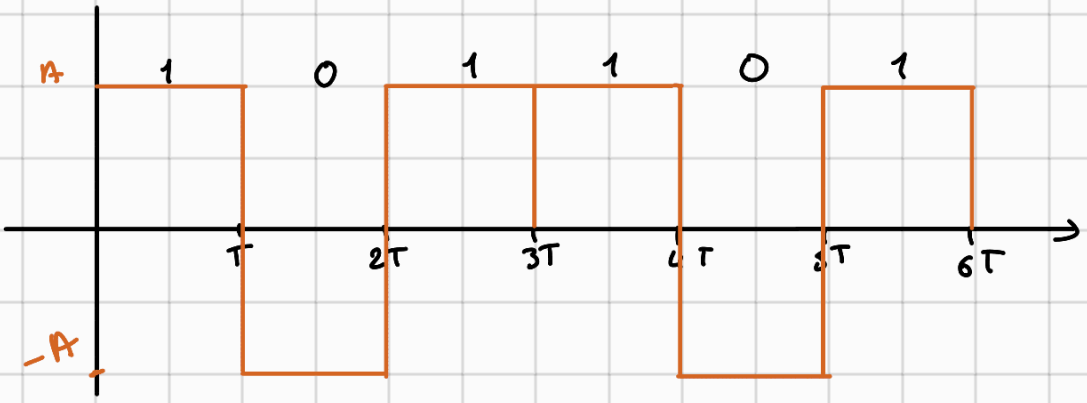
$$\sigma_0^2 = \frac{N_0}{2} E_h = \frac{N_0}{2} A^2 T$$

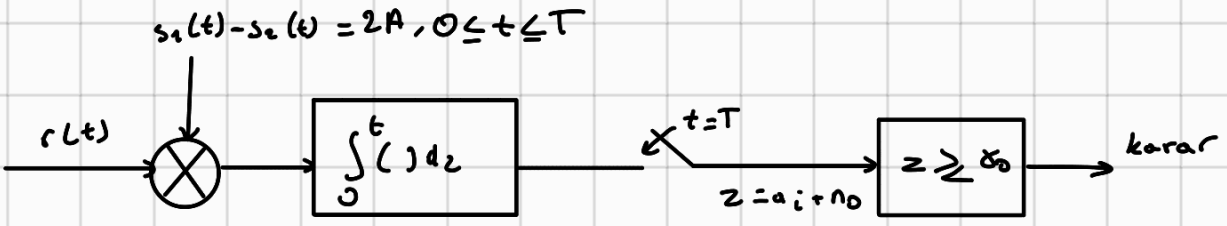
$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\frac{A^2 T - 0}{2\sqrt{\frac{N_0}{2} A^2 T}}\right) = Q\left(\sqrt{\frac{A^2 T}{2N_0}}\right) = Q\left(\sqrt{\frac{E_b}{N_0}}\right)$$

## Çift kutuplu işaretleşme
## (Bipolar signaling)

"1" → $s_1(t) = A,\ 0 \le t \le T$

"0" → $s_2(t) = -A,\ 0 \le t \le T$




$$a_i = \int_0^T s_i(t)\,\underbrace{[s_1(t) - s_2(t)]}_{2A}\,dt$$

$$a_1 = \int_0^T \underbrace{s_1(t)}_{A}\,2A\,dt = 2A^2T$$

$$a_2 = \int_0^T \underbrace{s_2(t)}_{-A}\,2A\,dt = -2A^2T$$

$$\gamma_0 = \frac{a_1 + a_2}{2} = 0$$

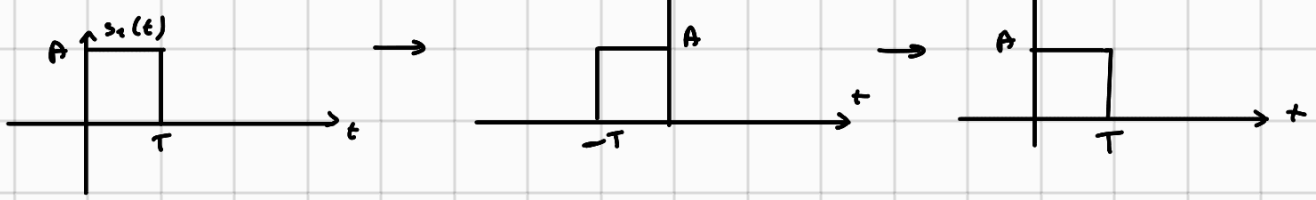
Eğer $z > 0 \Rightarrow$ karar ar: $s_1(t)$

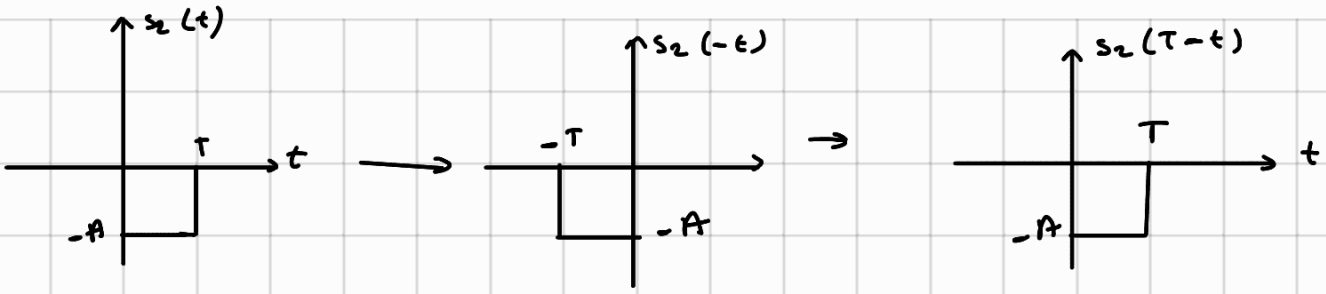
$z < 0 \Rightarrow$ " $= s_2(t)$

$$E_1 = E_2 = \int_0^T s_1^2(t)\,dt = A^2T \rightarrow E_b = \underbrace{E_1}_{A^2T}\,P(s_1) + \underbrace{E_2}_{A^2T}\,P(s_2) = A^2T$$

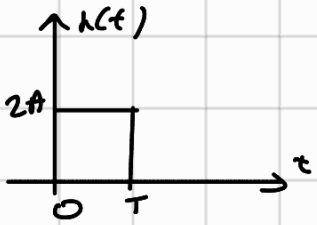
$P(s_1) = 1/2$, $P(s_2) = 1/2$

$E_n = ??$






$h(t) = 2A, \quad 0 \le t \le T$




$$E_h = \int_0^T h^2(t)\,dt = \int_0^T 4A^2\,dt = 4A^2T$$

$$\sigma_0^2 = \frac{N_0}{2} E_h = \frac{N_0}{2} 4A^2T = 2N_0A^2T$$

$$E_d = \int_0^T \underbrace{[s_1(t) - s_2(t)]^2}_{2A}\,dt = \int_0^T 4A^2\,dt = 4A^2T$$

$$P_b = Q\left(\frac{a_1 - a_2}{2\sigma_0}\right) = Q\left(\frac{2A^2T - (-2A^2T)}{2\sqrt{2N_0A^2T}}\right) = Q\left(\frac{4A^2T}{\sqrt{8N_0A^2T}}\right) = Q\left(\sqrt{\frac{2A^2T}{N_0}}\right)$$

$$= Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

$$P_b = Q\left(\sqrt{\frac{E_b}{N_0}(1-\rho)}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

($\rho$: -1)

$$P_b = Q\left(\sqrt{\frac{E_d}{2N_0}}\right) = Q\left(\sqrt{\frac{4A^2T}{2N_0}}\right) = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

Fig 3.14

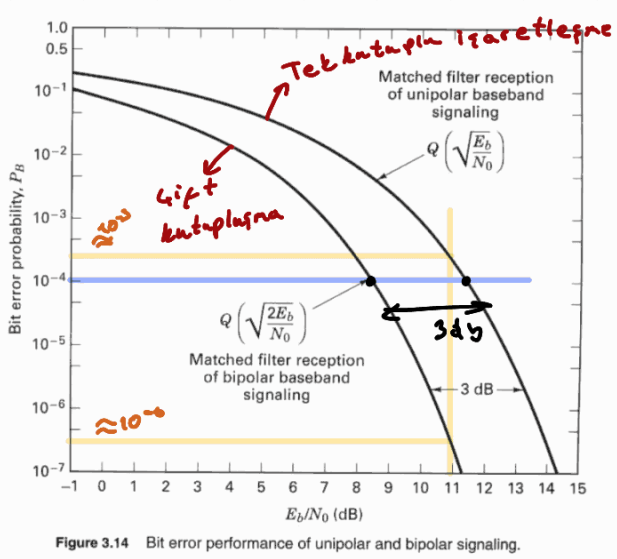



Figure 3.14 Bit error performance of unipolar and bipolar signaling.

$$10\log_{10}\frac{2E_b}{N_0} = \underbrace{10\log_{10}2}_{3\text{dB}} + 10\log_{10}\frac{E_b}{N_0}$$

Çift katmanlı daha iyi performans sağlar

Var gok haberlesmesi, ışık fibor haberleşmede kullanılır (tek kutuplu)

Örnek: Sklar, S 163, Prob35

**3.5.** A bipolar binary signal, $s_i(t)$, is a +1- or −1-V pulse during the interval $(0, T)$. Additive white Gaussian noise having two-sided power spectral density of $10^{-3}$ W/Hz is added to the signal. If the received signal is detected with a matched filter, determine the maximum bit rate that can be sent with a bit error probability of $P_B \le 10^{-3}$.

Max bit rate?

$R = \frac{1 \text{ bit}}{T_{sn}}$

$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) \le \underbrace{10^{-3}}_{0,001}$ $\qquad E_b = A^2T = 1^2T = T$

$\sqrt{\frac{2E_b}{N_0}} > 3,08$ (Tablodan)

$\Rightarrow \sqrt{\frac{2T}{N_0}} \ge 3,08 \Rightarrow \frac{2T}{N_0} \ge 9,4864 \Rightarrow T \ge \frac{N_0}{2} \cdot 9,47 = 9,48 \cdot 10^{-3}$ sn

$R = \frac{1}{T} \le 105,41$ bit/sn

Örnek Sklar, Prob3.14

**3.14.** Consider that NRZ binary pulses are transmitted along a cable that attenuates the signal power by 3 dB (from transmitter to receiver). The pulses are coherently detected at the receiver, and the data rate is 56 kbit/s. Assume Gaussian noise with $N_0 = 10^{-6}$ Watt/Hz. What is the minimum amount of power needed at the transmitter in order to maintain a bit-error probability of $P_B = 10^{-3}$?

NRZ-L, attenuation 3db

Coherent detection

$R = 56$ kbit/sec, $N_0 = 10^{-6}$ Watt/Hz

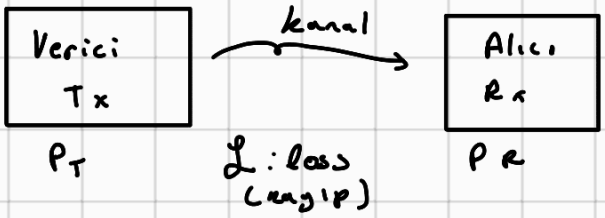
min amount of power needed at Tx for $P_b = 10^{-3}$

Bipolar signaling $\Rightarrow P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right) = 10^{-3}$

$E_b = A^2T$, $R = \frac{1}{T} = 56000 \frac{\text{bit}}{\text{sn}}$ $\Rightarrow Q\left(\sqrt{\frac{2A^2(1/56000)}{10^{-6}}}\right) = 10^{-3}$

$= 3,1$ tablodan

$\sqrt{\frac{2A^2(1/56000)}{10^{-6}}} = 3,1 \Rightarrow A^2 = 0,265$ Watt $\hat{=} 270$ mW




$$10\log_{10} P_T = 10\log_{10} d + 10\log_{10} P_R$$

$$\mathcal{L}_d = 10\log_{10}\frac{P_T}{P_R}$$

$P_R = A_r^2 = 270\ mW$

$P_T = P_R \cdot 10^{\mathcal{L}_{dB}/10}$

$= 270mW \cdot 10^{3/10}$

$= 540\ mW$

## Likelihood Ratio Test and Maximum a Posteriori (MAP) Criterion

(probability)

Muhtemellik oranı testi ve maksimum sonsal olasılık kriteri

$$P(s_1/z) \underset{H_2}{\overset{H_1}{\gtrless}} P(s_2/z)$$ (Hipotez)

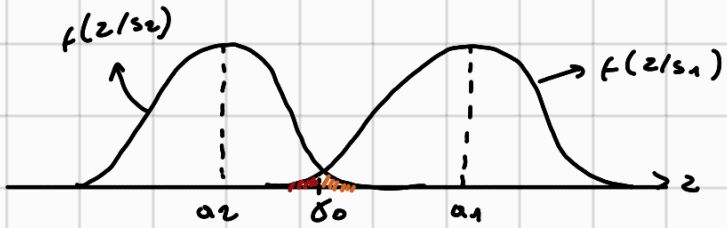
$$P(s_i/z) = \frac{f(z/s_i)\,P(s_i)}{f(z)} \qquad i=1,2$$

$$f(z) = \sum_{i=1}^{2} f(z/s_i)\,P(s_i) = f(z/s_1)\,P(s_1) + f(z/s_2)\,P(s_2)$$

$$f(z/s_1)\,P(s_1) \underset{H_2}{\overset{H_1}{\gtrless}} f(z/s_2)\,P(s_2)$$

$$\Lambda(z) = \frac{f(z/s_1)}{f(z/s_2)} \underset{}{\overset{H_1}{\gtrless}} \frac{P(s_2)}{P(s_1)}$$ likelihood ratio test

Eğer $P(s_1) = P(s_2) = \frac{1}{2} \Rightarrow \Lambda(z) = \frac{f(z/s_1)}{f(z/s_2)} \underset{H_2}{\overset{H_1}{\gtrless}} 1$ Maximum likelihood criterion (En muhteneli bulma kriteri)




$$\Lambda(z) = \frac{\frac{1}{\sqrt{2\pi}\,\delta_0} e^{-\frac{(z-a_1)^2}{2\delta_0^2}}}{\frac{1}{\sqrt{2\pi}\,\delta_0} e^{-\frac{(z-a_2)^2}{2\delta_0^2}}} \underset{H_2}{\overset{H_1}{\gtrless}} \frac{P(s_2)}{P(s_1)}$$

$$\Lambda(z) = \exp\left\{ -\frac{z^2+a_1^2-2za_1}{2\sigma_0^2} + \frac{z^2+a_2^2+2za_2}{2\sigma_0^2} \right\} \underset{H_2}{\overset{H_1}{\gtrless}} \frac{P(s_2)}{P(s_1)}$$

$$\frac{a_2^2-a_1^2}{2\sigma_0^2} + \frac{2z(a_1-a_2)}{2\sigma_0^2}$$

$$L(z) = \ln \Lambda(z) = \frac{a_2^2-a_1^2}{2\sigma_0^2} + \frac{z(a_1-a_2)}{\sigma_0^2} \underset{H_2}{\overset{H_1}{\gtrless}} \ln \frac{P(s_2)}{P(s_1)}$$

$$\frac{z(a_1-a_2)}{\sigma_0^2} \underset{H_2}{\overset{H_1}{\gtrless}} \ln \frac{P(s_2)}{P(s_1)} + \frac{a_1^2-a_2^2}{2\sigma_0^2}$$

$$z \underset{H_2}{\overset{H_1}{\gtrless}} \frac{\sigma_0^2}{(a_1-a_2)} \ln \frac{P(s_2)}{P(s_1)} + \frac{\sigma_0^2}{a_1-a_2} \frac{a_1^2-a_2^2}{2\sigma_0^2}$$

$a_1+a_2$ kalır

$$z \underset{H_2}{\overset{H_1}{\gtrless}} \frac{\sigma_0^2}{a_1-a_2} \ln \frac{P(s_2)}{P(s_1)} + \frac{a_1+a_2}{2} = \gamma_0$$

Eğer $P(s_1) = P(s_2) = \frac{1}{2} \Longrightarrow \gamma_0 = \frac{a_1+a_2}{2}$

$$P(h/s_1) = \int_{-\infty}^{\gamma_0} f(z/s_1)\, dz = \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{-\infty}^{\gamma_0} e^{-\frac{(z-a_1)^2}{2\sigma_0^2}} dz$$

$$= 1 - \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\gamma_0}^{\infty} e^{-\frac{(z-a_1)^2}{2\sigma_0^2}} dz = 1 - \frac{1}{\sqrt{2\pi}} \int_{\frac{\gamma_0-a_1}{\sigma_0}}^{} e^{-\frac{t^2}{2}} \sigma_0\, dt$$

$$Q\left(\frac{\gamma_0-a_1}{\sigma_0}\right)$$

$\frac{z-a_1}{\sigma_0} = t$ $\quad z=\gamma_0 \rightarrow t = \frac{\gamma_0-a_1}{\sigma_0}$ $\quad z=\infty \rightarrow t=\infty$

$\Rightarrow z = \sigma_0 t + a_1$

$dz = \sigma_0 \cdot dt$

$$\Rightarrow \boxed{P(h/s_1) = 1 - Q\left(\frac{\gamma_0-a_1}{\sigma_0}\right)}$$

$$P(h/s_2) = \int_{\gamma_0}^{\infty} f(z/s_2)\, dz$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\gamma_0}^{\infty} e^{-\frac{(z-a_2)^2}{2\sigma_0^2}} dz$$

$$= \frac{1}{\sqrt{2\pi}\,\sigma_0} \int_{\frac{\gamma_0-a_2}{\sigma_0}}^{\infty} e^{-\frac{t^2}{2}} \sigma_0\, dt$$

$$Q\left(\frac{\gamma_0-a_2}{\sigma_0}\right)$$

$$P(h/s_2) = Q\left(\frac{\gamma_0 - a_2}{\sigma_0}\right)$$

$$P_D = P(h/s_1) \cdot P(s_1) + P(h/s_2) \cdot P(s_2)$$

$$P_D = \underbrace{\left[1 - Q\left(\frac{\gamma_0 - a_1}{\sigma_0}\right)\right]}_{Q\left(\frac{a_1 - \gamma_0}{\sigma_0}\right)} P(s_1) + Q\left(\frac{\gamma_0 - a_2}{\sigma_0}\right) P(s_2)$$

$$Q(-x) = 1 - Q(x)$$

